

Simplifying expressions with rational exponents

The index laws applied to letters — and the factorising tricks that work just as well with fractional powers.

LESSON 32–33

2 × 40 MIN

ALGEBRA 8, §10

STAGE 9 · 2.3

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up 5 min

Explanation 12 min

Interactive 8 min

Practice 13 min

Homework 2 min

LEARNING OBJECTIVES

- Simplify products and quotients of powers with rational exponents.
- Take a common power out as a factor.
- Recognise a difference of squares in $a - b = (\sqrt{a})^2 - (\sqrt{b})^2$.
- Write an answer in root form when asked.

TEXTBOOK

National §10, pp. 49–52

Cambridge Learner's Book pp. 30–33

Workbook Workbook 2.3

01 Explanation

The routine

1. Turn every root into a power. Fractions of indices are easier than nested radicals.
2. Apply the index laws.
3. Collect, factorise or cancel.
4. Convert back to root form only if the question asks for it.

$$\frac{\sqrt{a} \cdot \sqrt[3]{a}}{\sqrt[6]{a}} = \frac{a^{1/2} \cdot a^{1/3}}{a^{1/6}} = a^{1/2 + 1/3 - 1/6} = a^{2/3}$$

Taking out a common power

The smallest index is the one to take out — exactly as with whole-number powers:

$$a^{3/2} + a^{1/2} = a^{1/2}(a + 1)$$

Check by multiplying back: $a^{1/2} \cdot a = a^{3/2}$ ✓ and $a^{1/2} \cdot 1 = a^{1/2}$ ✓.

The difference-of-squares trick

Since $a = (\sqrt{a})^2$ for $a \geq 0$, any difference of two non-negative numbers is a difference of squares:

$$a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$$

That is the single most useful move in this section. It cancels fractions such as

$$\frac{a - b}{\sqrt{a} - \sqrt{b}} = \sqrt{a} + \sqrt{b}$$

and it also rationalises a two-term denominator, by multiplying by the **conjugate**:

$$\frac{1}{\sqrt{5} - \sqrt{3}} = \frac{\sqrt{5} + \sqrt{3}}{(\sqrt{5})^2 - (\sqrt{3})^2} = \frac{\sqrt{5} + \sqrt{3}}{2}$$

02 Worked examples

EXAMPLE 1 Simplify $\frac{a^{3/4} \cdot a^{1/2}}{a^{1/4}}$

$$\textcircled{1} \quad \frac{3}{4} + \frac{1}{2} = \frac{5}{4}$$

Multiply on top: add the indices.

$$\textcircled{2} \quad \frac{5}{4} - \frac{1}{4} = 1$$

Divide: subtract the index.

$$\textcircled{3} \quad = a$$

ANSWER

a

EXAMPLE 2 Factorise $a^{3/2} - a^{1/2}$

① *smallest index is $\frac{1}{2}$*

That is the common factor.

② $a^{1/2}(a^1 - 1)$

Subtract $\frac{1}{2}$ from each index.

③ $= a^{1/2}(a - 1)$
or $\sqrt{a}(a - 1)$

ANSWER

$$\sqrt{a}(a - 1)$$

EXAMPLE 3 Simplify $\frac{x - 9}{\sqrt{x} - 3}$, $x \geq 0$, $x \neq 9$

① $x = (\sqrt{x})^2$ and $9 = 3^2$

A difference of squares in disguise.

② $x - 9 = (\sqrt{x} - 3)(\sqrt{x} + 3)$

③ $= \sqrt{x} + 3$

Cancel the bracket.

ANSWER

$$\sqrt{x} + 3$$

03 Interactive model

Use the model to check any answer numerically: put a value in and compare both forms.

Check an index answer numerically

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a¹ **64**

Raising the answer to the power 3 gives back a¹ — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Simplify	Soddalashtirish	Упростить
Common power	Umumiy daraja	Общая степень
Substitution	Almashtirish	Замена
Difference of squares	Kvadratlar ayirmasi	Разность квадратов
Factor out	Qavsdan chiqarish	Вынести за скобку
Root form	Ildiz ko‘rinishi	Форма с корнем
Index form	Daraja ko‘rinishi	Степенная форма
Expression	Ifoda	Выражение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself
 $a^{1/2} \cdot a^{1/2} \cdot a^{1/2}$ equals:
 $a^{3/2} + a^{1/2}$ factorises to:
 $\frac{x-4}{\sqrt{x}} - 2$ equals:

To rationalise $\frac{1}{\sqrt{5} - \sqrt{3}}$ multiply by:

$$\sqrt{5}$$

$$\sqrt{5} + \sqrt{3}$$

$$\sqrt{5} - \sqrt{3}$$

$$2$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify $a^{1/2} \cdot a^{1/2}$

ANSWER a

2. Simplify $a^{3/4} : a^{1/4}$

ANSWER $a^{1/2}$

3. Simplify $(a^{1/3})^3$

ANSWER a

4. Simplify $a^{1/2} \cdot a$

ANSWER $a^{3/2}$

5. Write $\sqrt{a} \cdot \sqrt{a}$ more simply

ANSWER a

6. Simplify $(4a^2)^{1/2}$, $a \geq 0$

ANSWER $2a$

7. Simplify $a^{1/2} + 3a^{1/2}$

ANSWER $4a^{1/2}$

Medium · the standard question

7 PROBLEMS

1. Simplify $\frac{a^{3/4} \cdot a^{1/2}}{a^{1/4}}$

ANSWER a

2. Factorise $a^{3/2} - a^{1/2}$

ANSWER $\sqrt{a}(a - 1)$

3. Simplify $\frac{x - 9}{\sqrt{x} - 3}$

ANSWER $\sqrt{x} + 3$

4. Simplify $(a^{1/2}b^{1/2})^4$

ANSWER a^2b^2

5. Simplify $\sqrt{a} \cdot \sqrt[3]{a}$

ANSWER $a^{5/6}$

6. Simplify $\frac{a}{\sqrt{a}}$, $a > 0$

ANSWER $\sqrt{a}$

7. Factorise $a + a^{1/2}$

ANSWER $a^{1/2}(a^{1/2} + 1)$

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\frac{\sqrt{a} \cdot \sqrt[3]{a}}{\sqrt[6]{a}}$

ANSWER $a^{2/3}$

2. Simplify $\frac{x-1}{\sqrt{x}-1}$, $x \geq 0, x \neq 1$

ANSWER $\sqrt{x} + 1$

3. Rationalise $\frac{1}{\sqrt{5}-\sqrt{3}}$

ANSWER $\frac{\sqrt{5}+\sqrt{3}}{2}$

4. Simplify $(a^{2/3} - b^{2/3}) : (a^{1/3} - b^{1/3})$

ANSWER $a^{1/3} + b^{1/3}$

5. Simplify $\frac{a^{3/2} - a^{1/2}}{a-1}$, $a > 0, a \neq 1$

ANSWER $a^{1/2}$

6. Evaluate $(\sqrt{7} + \sqrt{3})(\sqrt{7} - \sqrt{3})$

ANSWER $7 - 3 = 4$

7. Simplify $(a^{1/2} + 1)^2 - (a^{1/2} - 1)^2$

ANSWER $4a^{1/2}$

07 Homework

Homework — 5 problems

Algebra 8, §10, pp. 49–52.

1. Simplify $a^{2/3} \cdot a^{1/3}$

2. Simplify $\frac{b^{7/4}}{b^{3/4}}$

3. Factorise $x^{3/2} + x^{1/2}$

4. Simplify $\frac{x - 16}{\sqrt{x} - 4}$

5. Rationalise $\frac{2}{\sqrt{7} - \sqrt{5}}$